STAFF REVIEW · H446 267 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Read, interpret, execute an instruction.

2. Unavailable I/O or required event.

3. Which Ready process receives CPU time.

4. $A \wedge (B \vee \neg B) = A \wedge 1 = A$.

I Do — source worked steps

1. Predict first — The ISR processes keyboard input before restoring the prior program state.

We Do — guided CHECK

1. Predict first
The ISR can acknowledge completion and allow queued device work to proceed; the interrupted task resumes.

You Do — source mark schemes

1. Interrupt signals attention [1], ISR is code handling event [2], e.g. keyboard or timer [1].

2. Interrupt arrives [1]; preserve PC/required registers [2]; ISR handles event [1]; restore [1]; correct original program continuation [1].

3. Polling repeatedly checks [2], wastes cycles [1]; interrupt-driven signals events [1], frees CPU for other work [1], adds ISR/context overhead [1].

Exit Ticket — source answer

1. False. Necessary state must be saved and restored.